

We Are Different And We Are Similar

Universals, Commonality, Diversity and Variation In Parenting and Child Development

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1 Acknowledgements

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2 The Big Ideas

2.1 The Scope of the Issue

- UNICEF considers children globally to be in crisis: political violence, climate change, and poverty affect many children (UNICEF, 2024a, 2024b).
- In this time of crisis, parenting is one crucial area of focus.
- According to UNICEF data, $\frac{1}{3}$ of children are subject to physical punishment (Ward et al., 2023); $\frac{2}{3}$ to some form of violent punishment (UNICEF, 2024a).
- In a review of 50 years of literature on physical punishment, Gershoff & Grogan-Kaylor (2016) connect physical punishment to poor mental health outcomes, and other poor outcomes over the life course. Ward et al. (2023) examines the relationship of multiple forms of parental discipline to child outcomes.

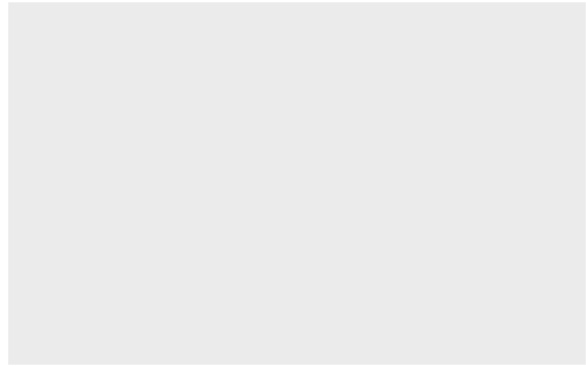
2.2 Are We Similar? Are We Different?

2.3 Both Universals and Variation Are Important

“Despite the incredible diversity existing among and within human cultures, there are many phenomena that occur regularly in all known societies. These commonalities,



(a) positive parenting



(a) countries in MICS data

or universals, while deriving in part from human nature, may also have specific social, cultural, and systemic sources. We need to develop a working understanding of these universals so that we might advance legitimate, empirically based human science set on creating knowledge that is politically relevant to fostering real solutions to the problems that complicate human co-existence in the Age of the Anthropocene.” (Antweiler, 2016)

2.4 Highlighting Commonalities May Promote Intergroup Attitudes And Solidarity

“Hanel et al. (2019) cite research suggesting that ‘highlighting similarities between groups improves interpersonal and intergroup attitudes.’ Others have argued that ‘build[ing] bonds of commonality across our differences’ might be an impetus for social transformation (Hunt-Hendrix & Taylor, 2024).”

2.5 The Argument Of This Presentation

- Families, parents and children experience an incredible diversity of circumstances.
- Parents engage in parenting behaviors with consequent diversity, though with some surprising similarities.
- The associations of parenting with child outcomes are surprisingly consistent.

3 The Study

3.1 Data

MICS data collected by UNICEF from Low and Middle Income Countries (LMIC) around the world. $N = 224,632$ families; $N_{countries} = 60$

Contains questions about parental discipline (both positive and negative) and child aggression .

```
Error in `value[[3L]]()`:  
! Failed to install the rnaturalearthdata package. Please try installing  
  the package for yourself using the following command:  
  `install.packages("rnaturalearthdata")`
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! object 'mapdata' not found
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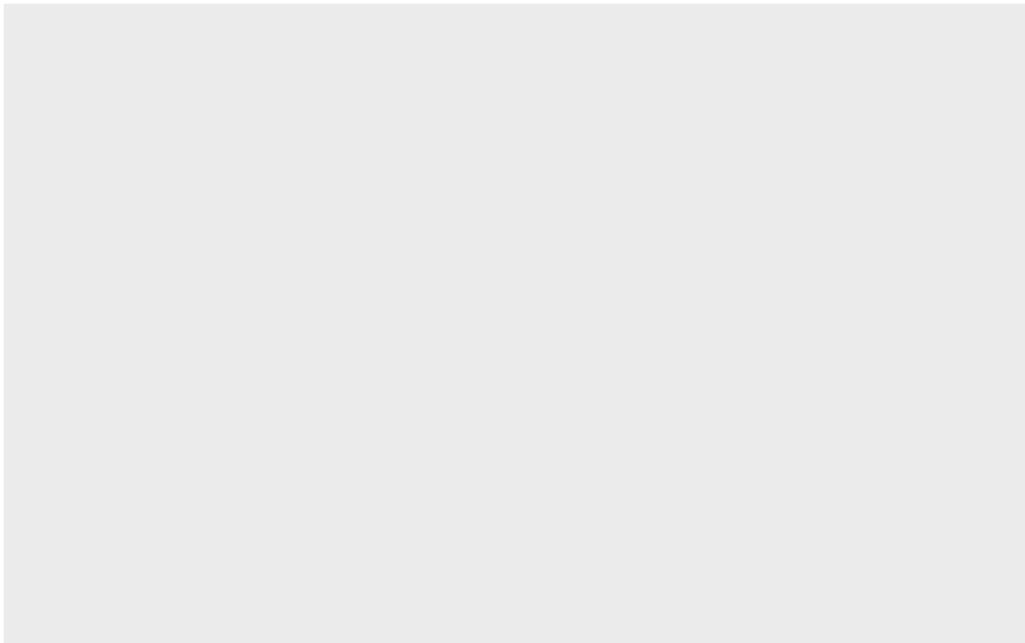
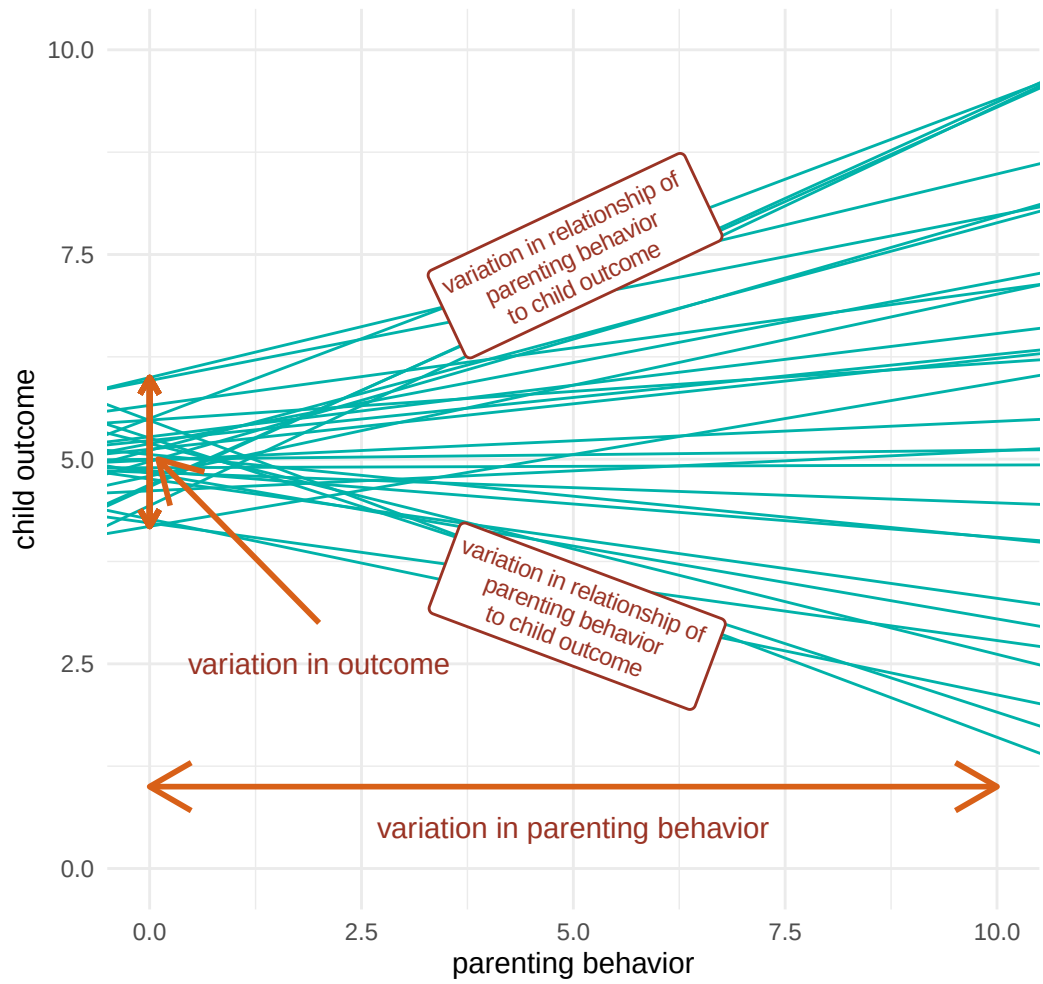


Figure 3: countries in UNICEF MICS Data

Sources of Variation

In The Study of Parenting And Child Outcomes



This kind of graph is called a SPAGHETTI PLOT.
Every line is a country

Figure 4: Sources of Variation In The Study of Parenting And Child Outcomes

3.2 Sources of Variation In The Study of Parenting And Child Outcomes

3.3 Check-In

Are we OK? Are there questions?

3.4 Sources of Variation (2)

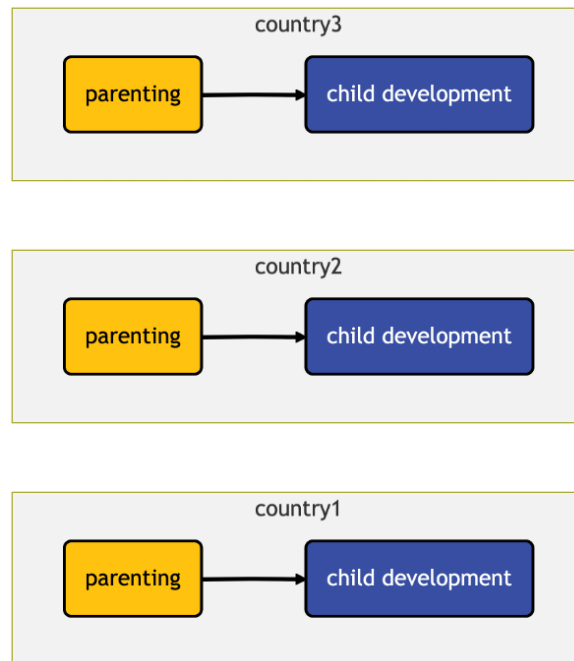


Figure 5: a visual version of the research question

3.5 The Equation

$$\text{outcome} = \beta_0 + \Sigma \beta \text{parenting} + \Sigma \beta \text{covariates} + u_{0j} + \Sigma u_{kj} \times \text{parenting}$$

Annotations for the equation:

- baseline** points to β_0 .
- linear or dichotomous outcome** points to outcome .
- parenting** points to $\Sigma \beta \text{parenting}$.
- covariates** points to $\Sigma \beta \text{covariates}$.
- country level variation in outcome** points to u_{0j} .
- country level variation in associations** points to $\Sigma u_{kj} \times \text{parenting}$.

Figure 6: annotated equation

3.6 Parenting Behaviors Vary *Somewhat* Across Countries

Verbal reasoning and *shouting* are the most common parental discipline behaviors toward young children.

Proportions of Parents Using Different Discipline Types

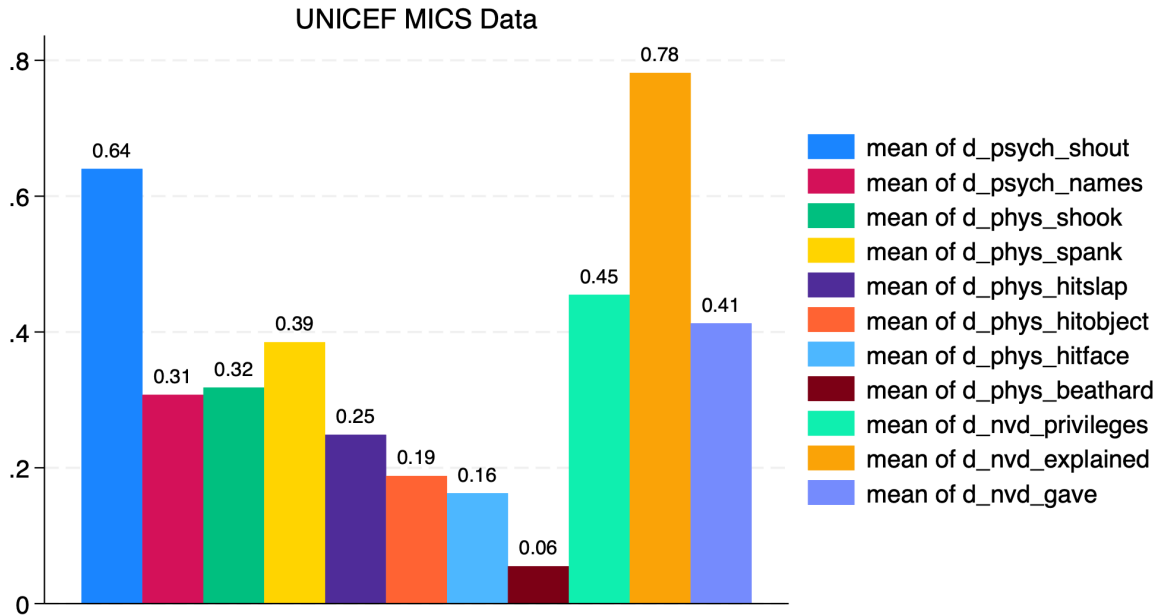


Figure 7: positive and negative parenting

3.7 The *Maximal Model* Approach

- Hypothetically, one might imagine that there could be group level unobserved factors which affect many regression slopes—i.e. the relationship between multiple predictors (e.g. x_1 , x_2 , x_3 , etc.) and outcome variable y .
- Arguably, were one to ignore these unobserved factors in statistical estimation, they would show up either in an error term (e_i or u_0), or in the regression coefficients (β). This could be seen as a form of statistical bias.
- The above could lead to a substantive mis-estimation of important effects. Thus, there is a conceptual argument for including as many random effects—i.e. random slopes—in a statistical model as possible (Barr et al., 2013; Frank, 2018).
- There is also a substantive reason: We might be interested in the *variation* of *all* the random slopes!

Sources of Variation In The Study of Parenting And Child Outcomes

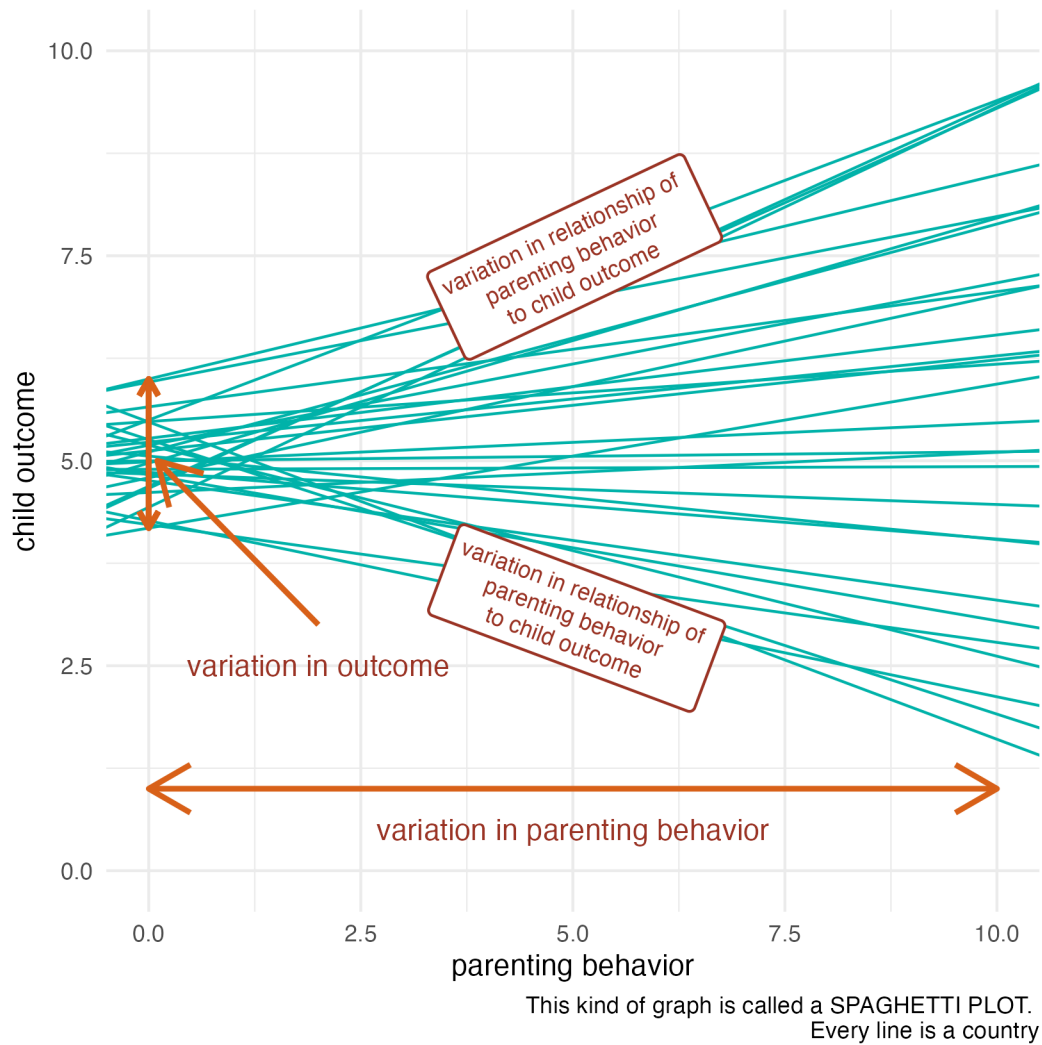


Figure 8: sources of variation

3.8 Relationship of Parenting And Child Outcomes

Maximal Model of Parenting and Child Aggression Outcome

	aggress	
child aggression		
Sex of randomly selected child	-0.023	**

	aggress	
Age of randomly selected child	0.000	
Selected child shouted at	0.036	**
Selected child called dumb/lazy/names	0.054	**
Selected child shaken	0.039	**
Selected child spanked	0.023	**
Selected child hit/slapped on hand/arm/leg	0.033	**
Selected child hit with belt/brush/stick/etc	0.035	**
Selected child hit/slapped on face/head/ears	0.044	**
Selected child beat as hard as one could	0.026	**
Selected child's privileges taken away	0.007	
Selected child - explained why behavior wrong	-0.014	*
Selected child given something else to do	0.004	
Intercept	0.325	**
country		
var(d_psych_shout)	0.001	
var(d_psych_names)	0.001	
var(d_phys_shook)	0.001	
var(d_phys_spank)	0.001	
var(d_phys_hitslap)	0.001	
var(d_phys_hitobject)	0.001	
var(d_phys_hitface)	0.002	
var(d_phys_beathard)	0.002	
var(d_nvd_privileges)	0.001	
var(d_nvd_explained)	0.001	
var(d_nvd_gave)	0.001	
var(__cons)	0.012	
Residual		
var(e)	0.217	
Number of observations	224632	

** p<.01, * p<.05

4 The Relationship of Parenting And Child Outcomes Shows Little Variation Across Countries

4.1 A Standard Spaghetti Plot

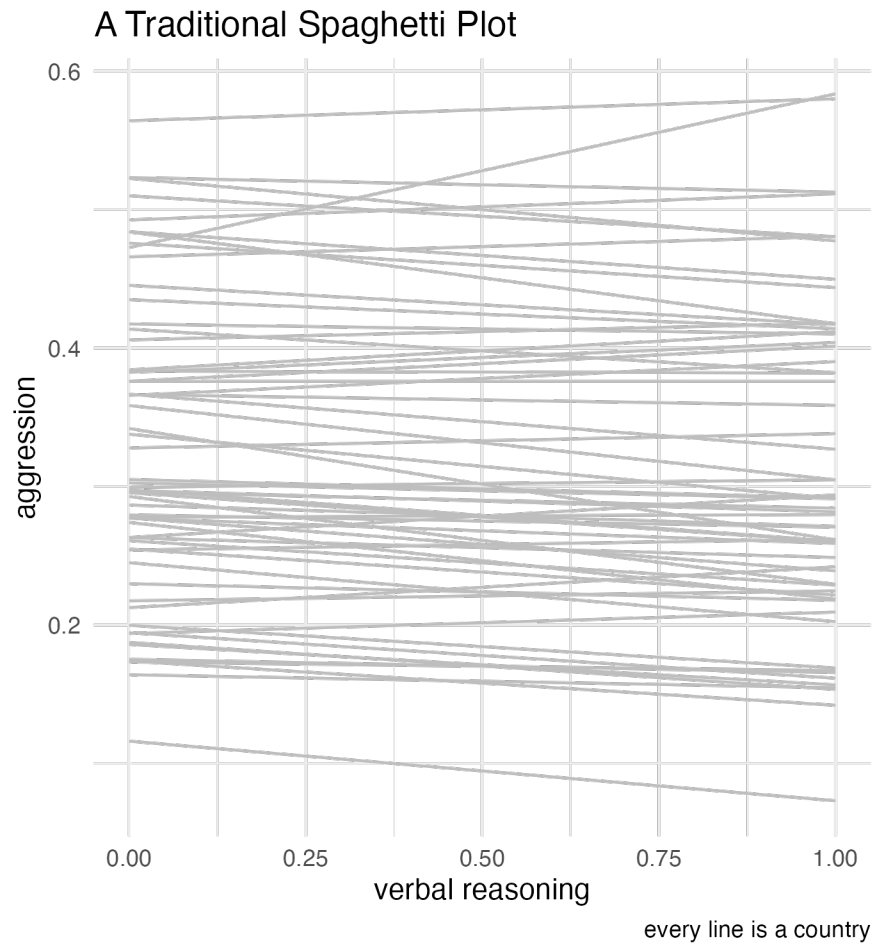


Figure 9: a standard 'spaghetti' plot

4.2 A Modified Spaghetti Plot

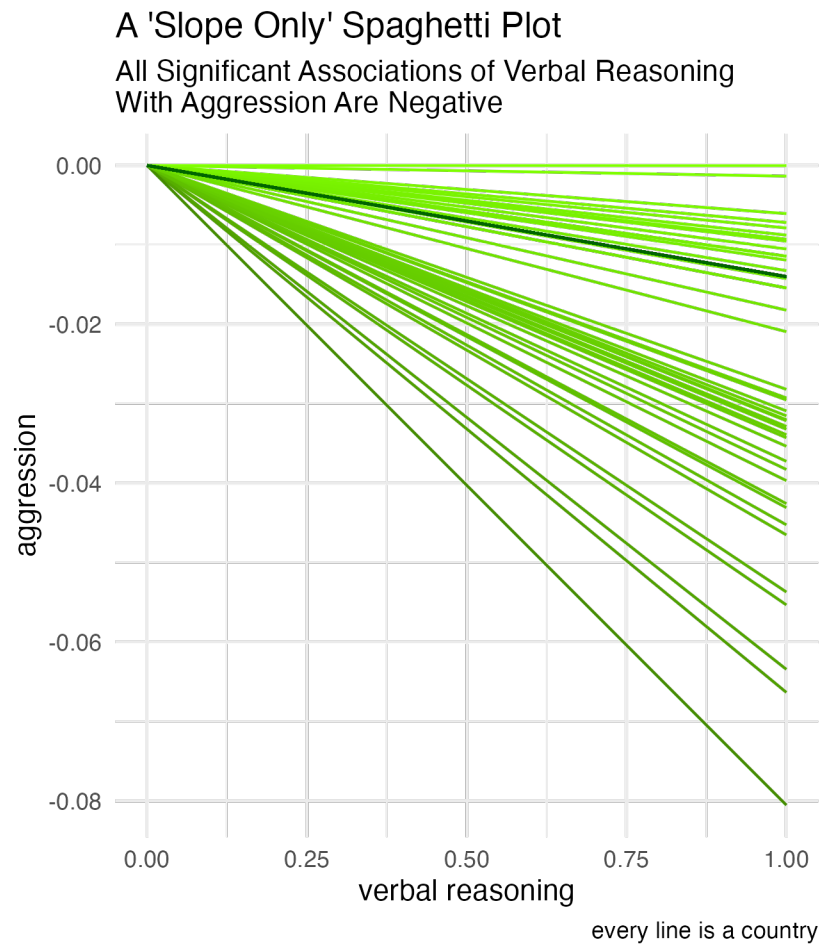


Figure 10: modified spaghetti plot

4.3 Quantifying Diversity and Commonality in the Relationship of Parenting To Child Outcomes

1. Regression slopes—when statistically significant—are all in the same direction, and of at least somewhat similar magnitude.
2. The actual values of the $\text{var}(u_{kj})$ or $\text{range}(u_{kj})$ are small.

5 Wrapping It Up

5.1 To Summarize ...

- There is an incredible diversity across countries, especially LMIC's.
- However, the associations of different forms of parental discipline are surprisingly consistent.
 - Explaining to children the difference between right and wrong has consistently “good” outcomes in terms of aggression.
 - Spanking has consistently “bad” outcomes in terms of aggression.
 - Shouting has consistently “bad” outcomes in terms of aggression.
 - Name calling has consistently “bad” outcomes in terms of aggression.

5.2 Attempting a Visual Summary...

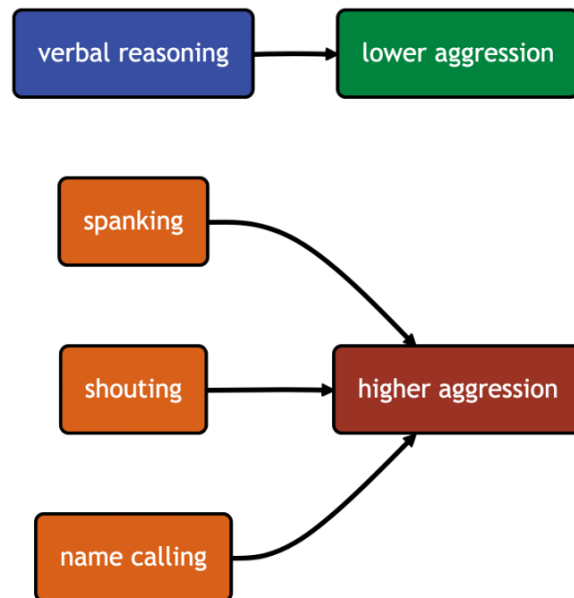


Figure 11: a visual summary

5.3 Implications

“My conception of the universal is that of a universal enriched by all that is particular, a universal enriched by every particular: the deepening and coexistence of all particulars.” (Césaire, 1956)

- Associations of parental discipline with child outcomes are very consistent across countries.
- This may be helpful to international organizations (e.g. UNICEF, WHO) that are developing programs or interventions that are to be applied in multiple cultural settings.
- While cultural tailoring will always be necessary to some degree; child development research can inform the foundations of widely applicable interventions.

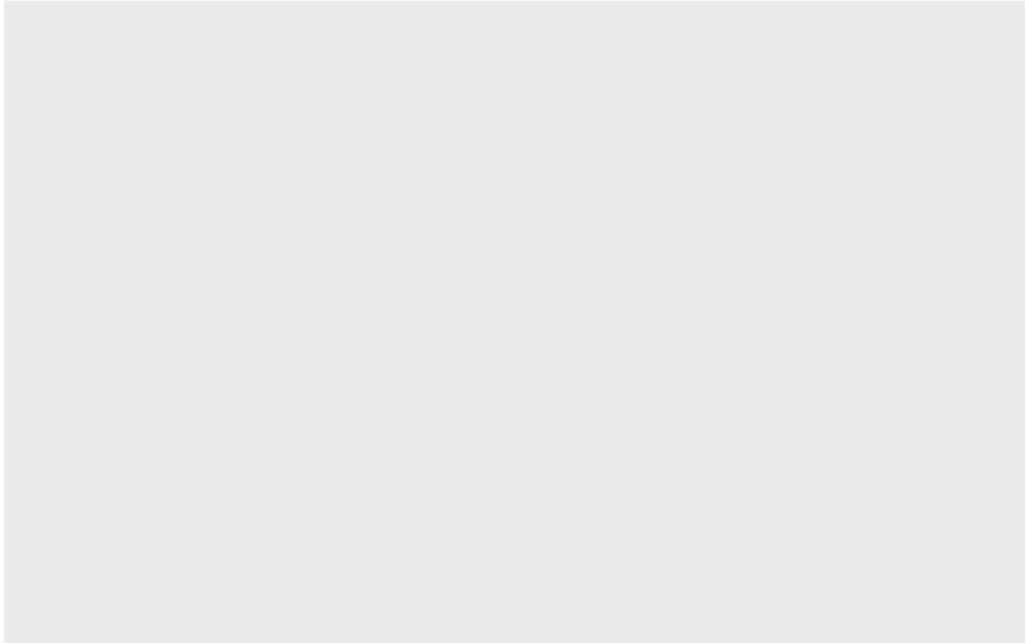


Figure 12: countries in MICS data

5.4 Discussion and Questions

How might it be helpful to start thinking about possible universality, commonality, and similarity ...

... while keeping in mind ...

... our diversity, differences, and variation?

5.4.1 Other Questions?

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